A global strict Lyapunov obstruction for RKCD

Abstract

We give an explicit obstruction to global strict state Lyapunov descent for one prescribed Runge–Kutta–Chebyshev descent map. The damping is $\eta = 1/20$, the global curvature bounds are 1 and 145, and the original parameter rule selects two stages and step 162/3361. The fixed analytic potential $f(x) = x^2/2 + 144 \log\cosh x$ satisfies these curvature bounds everywhere. Exact stage inequalities produce an admissible prime two-cycle and also show that the original potential increases from the initial state 3. Consequently, no finite real-valued state function decreases strictly at every nonzero point. Nevertheless, the same coefficients contract every quadratic whose Hessian spectrum lies in $[1, 145]$. The method, its quadratic certificate and the general spurious-cycle mechanism are inherited. The purpose is to verify the limitation for this literal prescription; no attracting basin, maximal range or firstness is asserted.

Keywords: Lyapunov function; discrete gradient dynamics; stabilized iteration; strong convexity; periodic orbit

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1 Introduction

A strict Lyapunov function can turn a discrete energy estimate into an actual trajectory conclusion. Before seeking such a function globally, it is useful to test the recurrence itself. A nonconstant periodic orbit excludes strict state decrease away from the proposed limiting equilibrium. This elementary obstruction is stronger than the failure of a particular candidate energy or a pairwise contraction estimate.

The RKCD method of Eftekhari, Vandereycken, Vilmart and Zygalakis [1] applies a stabilized explicit Runge–Kutta integrator to a gradient system. Their quadratic analysis gives a strict contraction on a prescribed curvature interval. A separate theorem treats a partitioned iteration, while general nonquadratic behavior is illustrated numerically. We examine the original full-gradient Algorithm 1 at one of its permitted positive dampings. The resulting example concerns the existence of a global strict state energy and convergence from every original state.

Spurious equilibria and cycles of Runge–Kutta methods have a substantial history; see Hairer, Iserles and Sanz-Serna [2] and Humphries [3]. Humphries and Stuart [4] study Lyapunov preservation and long-time behavior at sufficiently small steps. Modern cycle constructions also complement the search for Lyapunov functions in first-order methods [5]. We use only elementary real-variable estimates. The contribution considered here is the explicit realization for the stated RKCD prescription, with global analytic strong convexity and global smoothness.

We first specify the source algorithm. Let $\mathcal{C}_j$ denote the first-kind Chebyshev polynomial, so that

$$\mathcal{C}_0(t) = 1, \qquad \mathcal{C}_1(t) = t, \qquad \mathcal{C}_{j+1}(t) = 2t\mathcal{C}_j(t) - \mathcal{C}_{j-1}(t).$$

Given declared curvature bounds $0 < \ell < L$ and damping $\eta > 0$, Algorithm 1 and equation (3.6) of [1] prescribe

$$\begin{aligned} \kappa = L/\ell, \qquad s = \left\lceil \sqrt{(\kappa-1)\eta/2} \right\rceil, \qquad \omega_0 = 1 + \eta/s^2, \\ \omega_1 = \frac{\mathcal{C}_s(\omega_0)}{\mathcal{C}_s'(\omega_0)}, \qquad h = \frac{\omega_0 - 1}{\omega_1 \ell}, \qquad \mu_1 = \frac{\omega_1}{\omega_0}, \\ \mu_j = \frac{2\omega_1 \mathcal{C}_{j-1}(\omega_0)}{\mathcal{C}_j(\omega_0)}, \qquad \nu_j = \frac{2\omega_0 \mathcal{C}_{j-1}(\omega_0)}{\mathcal{C}_j(\omega_0)} \quad (2 \le j \le s). \end{aligned} \tag{1}$$

For a scalar potential f, each macro-step from x resets its stages by

$$\begin{aligned} X_0 &= x, \qquad X_1 = x - h\mu_1 f'(x), \\ X_j &= -h\mu_j f'(X_{j-1}) + \nu_j X_{j-1} - (\nu_j - 1) X_{j-2} \quad (2 \le j \le s), \\ T(x) &= X_s. \end{aligned} \tag{2}$$

There is one original state x_0 and the trajectory is $x_{n+1} = T(x_n)$. Internal stages have no independent initialization. In finite dimension the same definition uses ∇f.

Throughout the rest of this note, fix

$$\ell = 1, \qquad L = 145, \qquad \eta = \frac{1}{20}, \qquad f(x) = \frac{x^2}{2} + 144 \log\cosh x. \tag{3}$$

The positive damping convention in [1] permits this value. The number 1.17 displayed there is an example and is not imposed in (1).

Theorem 1. *For* (1)–(3)*, the following assertions hold.*

(i) *The potential is real analytic on* $\mathbb{R}$*, coercive and* 1*-strongly convex, with unique minimizer* 0 *and*

$$f(0) = f'(0) = 0, \qquad f(x) \ge x^2/2, \qquad 1 \le f''(x) \le 145 \quad (x \in \mathbb{R}). \tag{4}$$

The parameter rule selects $s = 2$ *and* $h = 162/3361$*. Writing* $g = f'$*, the literal map is*

$$Y(x) = x - \frac{g(x)}{81}, \qquad T(x) = x - \frac{81}{3361}\big(g(x) + g(Y(x))\big). \tag{5}$$

(ii) *The set* $\{x \in [3,4] : T(x) + x = 0\}$ *is nonempty and compact. If* r *is its smallest member, then* $3 < r < 4$ *and the admissible original start* $x_0 = r$ *gives*

$$x_{2n} = r, \qquad x_{2n+1} = -r \qquad (n \ge 0). \tag{6}$$

Its prime period is two, its distance from the minimizer is constantly r, and its potential is constantly $f(r) > 0$. Moreover, $f(T(3)) > f(3)$. There is no finite real-valued function $V : \mathbb{R} \to \mathbb{R}$ satisfying

$$V(T(x)) < V(x) \qquad \text{for every } x \neq 0. \tag{7}$$

(iii) *Retain the same declared bounds and coefficients for every comparison quadratic. Its scalar error multiplier at curvature $\lambda \in [1, 145]$ is*

$$R(\lambda) = \frac{(\lambda - 81)^2 - 3200}{3361}, \qquad \max_{1 \le \lambda \le 145} |R(\lambda)| = \frac{3200}{3361} < 1. \tag{8}$$

For every positive-definite quadratic in finite dimension whose Hessian spectrum lies in $[1, 145]$, one macro-step contracts Euclidean error by at most $3200/3361$ and objective error by at most its square.

The theorem concerns $\eta = 1/20$ and the resulting fixed map. It makes no statement about the illustrative choice $\eta = 1.17$, an attracting or repelling cycle, a basin, or the behavior of every other start. The obstruction in (7) is to an autonomous state function with strict decrease at every nonzero point. Nonstrict, time-dependent, local and suitably restricted-domain energies are outside this assertion.

2 Proof of Theorem 1

We first verify the prescribed map and the global objective. Then we prove two exact stage inequalities; these supply the entire infinite orbit without computing a numerical root.

Lemma 1. *For (3), the source coefficients are*

$$\begin{gathered} s = 2, \qquad \omega_0 = \frac{81}{80}, \qquad \omega_1 = \frac{3361}{12960}, \qquad h = \frac{162}{3361}, \\ \mu_1 = \frac{3361}{13122}, \qquad \mu_2 = \frac{1}{2}, \qquad \nu_2 = \frac{6561}{3361}. \end{gathered} \tag{9}$$

The potential satisfies (4), and the maps Y, T in (5) are globally defined continuous odd real-analytic maps.

Proof. The radicand in the stage rule is $(145 - 1)/(2 \cdot 20) = 18/5$, strictly between 1 and 4. Thus $s = 2$. Since $\mathcal{C}_2(t) = 2t^2 - 1$,

$$\mathcal{C}_2(81/80) = \frac{3361}{3200}, \qquad \mathcal{C}_2'(81/80) = \frac{81}{20}.$$

Substitution gives (9). In particular,

$$h\mu_1 = \frac{1}{81}, \qquad h\mu_2 = \nu_2 h\mu_1 = \frac{81}{3361}, \qquad \nu_2 - 1 = \frac{3200}{3361}.$$

Using these values in both stages of (2) gives precisely (5), including its old-stage term.

Because $\cosh x > 0$ on $\mathbb{R}$, composition with the real-analytic logarithm shows that f is real analytic at every real point. Direct differentiation gives

$$g(x) = x + 144 \tanh x, \qquad f''(x) = 1 + 144 \operatorname{sech}^2 x. \tag{10}$$

Thus $1 \le f'' \le 145$ globally. The upper endpoint is attained at zero and the lower infimum is approached as $|x| \to \infty$. Integration of the lower Hessian bound proves 1-strong convexity; the upper bound proves that g is globally 145-Lipschitz. Also $\log\cosh x \ge 0$, so $f(x) \ge x^2/2$, with $f(0) = g(0) = 0$. Hence f is coercive and zero is its unique minimizer. It is even, g is odd, and (5) gives the remaining assertions. Real analyticity here is a real-variable statement; no entire complex continuation is required. ∎

Lemma 2. *The actual finite stages of* (5) *satisfy*

$$T(3) + 3 < -\frac{585}{3361} < 0, \qquad T(4) + 4 \ge \frac{2916}{3361} > 0. \tag{11}$$

Proof. Write $t(x) = \tanh x$, $u(x) = \tanh Y(x)$ and $\Phi(x) = T(x) + x$. Exact substitution in (5) gives

$$Y(x) = \frac{80x - 144t(x)}{81}, \qquad 3361\Phi(x) = 6561x - 11520t(x) - 11664u(x). \tag{12}$$

The second hyperbolic tangent is evaluated at the actual first stage.

For a positive argument the exponential series has positive terms. In particular,

$$e^6 > 1 + 6 + \frac{6^2}{2} + \frac{6^3}{6} = 61 > 39.$$

Since $(v-1)/(v+1)$ is increasing for $v > 0$, the identity $\tanh a = (e^{2a} - 1)/(e^{2a} + 1)$ implies $\tanh 3 > 19/20$. On the other hand $\tanh 3 < 1$, so

$$Y(3) > \frac{240 - 144}{81} = \frac{32}{27} > \frac{7}{6}.$$

Another rational lower bound is

$$e^{7/3} > \sum_{j=0}^{4} \frac{(7/3)^j}{j!} = \frac{18289}{1944} = 9 + \frac{793}{1944} > 9.$$

It follows that $\tanh Y(3) > 4/5$. Equation (12) now yields

$$3361\Phi(3) < 19683 - 11520\frac{19}{20} - 11664\frac{4}{5} = -\frac{2961}{5} < -585.$$

At $x = 4$, both $\tanh 4$ and $\tanh Y(4)$ are strictly less than 1. They are finite because the map is globally defined. Consequently,

$$3361\Phi(4) > 26244 - 11520 - 11664 = 3060 > 2916.$$

These stronger inequalities imply (11). No numerical evaluation of a stage, exponential, hyperbolic tangent or root has been used. ∎

Proof of Theorem 1. Lemma 1 proves part (i). The function $\Phi = T + \mathrm{id}$ is continuous, and Lemma 2 gives opposite signs at 3 and 4. Hence

$$Z_0 = \{x \in [3,4] : \Phi(x) = 0\}$$

is nonempty. It is closed in a compact interval, and therefore its smallest member r exists. Neither endpoint is a root, so $3 < r < 4$. This defines one fixed exact initial state; uniqueness of the root is not asserted.

By definition $T(r) = -r$, and oddness gives $T(-r) = r$. At a macro-state σr, with $\sigma \in \{1, -1\}$, the actual reset stages are

$$X_0 = \sigma r, \qquad X_1 = Y(\sigma r) = \sigma Y(r), \qquad X_2 = T(\sigma r) = -\sigma r.$$

Induction proves (6) for every $n \geq 0$. Since $r > 0$, the two states differ and the prime period is two. Their distance from the unique minimizer is r, and evenness gives the constant positive objective $f(r) \geq r^2/2 > 0$. The even and odd subsequences have distinct limits, so the trajectory does not converge.

Separately, (11) gives $T(3) < -3$. Since $g(x) > 0$ for $x > 0$, evenness and strict increase of f on the positive half-line imply

$$f(T(3)) = f(|T(3)|) > f(3).$$

The energy rise thus holds at the original state 3; it is not being inferred from the periodic initialization. If a finite real-valued V satisfied (7), then the two cycle states would give

$$V(-r) < V(r) \quad \text{and} \quad V(r) < V(-r),$$

a contradiction. No continuity, positivity or coercivity assumption on V is needed for this obstruction. This proves part (ii).

For a scalar quadratic of curvature λ, the fixed stages act on the error as

$$Y = (1 - \lambda/81)x, \qquad T(x) = \left(1 - \frac{162\lambda}{3361} + \frac{\lambda^2}{3361}\right) x.$$

This is (8). On the whole interval $\lambda \in [1, 145]$, the quantity $\lambda - 81$ ranges over $[-80, 64]$, and its square ranges over $[0, 6400]$. The range of R is consequently

$$\left[-\frac{3200}{3361}, \frac{3200}{3361}\right].$$

Both endpoints are attained, respectively at $\lambda = 81$ and $\lambda = 1$.

Finally, let $q(x) = x^{\mathsf{T}}Ax/2 - b^{\mathsf{T}}x + c$, with symmetric positive-definite A whose spectrum is in $[1, 145]$. Relative to $x_* = A^{-1}b$, the same prescribed map has error update $e^+ = R(A)e$. Orthogonal diagonalization gives

$$\|e^+\|_2 \le \frac{3200}{3361}\|e\|_2, \qquad q(x^+) - q(x_*) \le \left(\frac{3200}{3361}\right)^2 \big(q(x) - q(x_*)\big).$$

The coefficients were held fixed throughout the interval; no tighter bounds were used to retune each quadratic. Part (iii) follows. ∎

3 Scope of the comparison

The source's Proposition 1 and the RKCD estimate (3.7) in [1] concern positive-definite quadratics. Its Proposition 3 concerns PRKCD, a different map which holds the nonlinear gradient component fixed at the macro-state during the internal stages. Theorem 1 is compatible with both results. It addresses a global nonlinear strict-energy extrapolation at the exact positive damping and declared curvature bounds in (3).

The small-step results of Humphries [3] and Humphries–Stuart [4] do not assert stability at every step allowed by a quadratic stability polynomial. Likewise, the classical RKC analysis of Verwer, Hundsdorfer and Sommeijer [6] proves linear finite-time error estimates. The newer work of Almuslimani, Vilmart and Zygalakis [7] repeats the quadratic RKCD guarantee and analyzes partitioned inner solvers for linear problems; it is not a global nonlinear theorem for the present map.

Pairwise contractivity is another distinct issue. Sanz-Serna and Zygalakis [8] construct convex-gradient examples where an RK method is not uniformly contractive between two starts. Such a one-step expansion does not by itself establish a nonconvergent trajectory or the failure of the original potential to decrease. Here both the energy rise and the complete periodic orbit are checked directly.

The general possibility of spurious numerical cycles and their incompatibility with strict state decrease are classical [2, 3, 5]. We make no firstness or exhaustive-priority claim for that mechanism. The present potential is fixed, analytic and globally strongly convex with globally Lipschitz gradient, and its cycle uses the source's automatic finite-stage prescription. No attraction or basin of the cycle is established. The example neither determines a maximal step range nor analyzes $\eta = 1.17$. The accompanying technical note records the source versions and access limits of the bounded comparison.

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